

Student Name: Humna Imran

TASK # 1: PYTHON

PANDAS DATA

CLEANUP

Imagine you have a messy notebook full of homework answers. Some answers are missing, some are written in different handwriting, and some dates are squished together. This document explains, block by block, how our code cleans that messy notebook step by step.

Block 1: Importing Tools

```
import pandas as pd
import numpy as np
import re
```

In simple words: This is like opening your pencil case before starting homework. We're just getting our tools (pandas, numpy, re) ready to use.

Dry run: Nothing shows up on screen yet, the tools are just loaded and ready to go.

Block 2: Creating a Practice Dataset

```
data = { ' Customer ID ': range(1001, 1201), ... }
raw_df = pd.DataFrame(data)
raw_df.to_csv('raw_data.csv', index=False)
```

In simple words: Since we don't have a real file, we make a pretend messy table ourselves, kind of like writing a practice test before the real one.

Dry run: A table with 200 rows is created, with some blanks left on purpose, like a worksheet with a few missing answers.

Block 3: Loading the CSV File

```
df = pd.read_csv('raw_data.csv')
df.head()
```

Student Name: Humna Imran

In simple words: This opens the file and brings the table into the computer's memory, like opening your notebook to start reading it.

Dry run: The first 5 rows show up on screen, something like columns for Customer ID, Customer Name, and Region, with real and missing values mixed in.

Block 4: Checking How Messy the Data Is

```
df.isnull().sum()
```

In simple words: This counts how many blank boxes are in each column, just like a teacher counting how many questions a student left unanswered.

Dry run: Output might say customer_name: 9, meaning 9 names are missing out of 200.

Block 5: Cleaning Up Column Names

```
df.columns = [normalize_column_name(c) for c in df.columns]
```

In simple words: Column titles like " Customer ID " (with extra spaces and capital letters) become customer_id, neat and easy to read.

Dry run: ' Order Amount (PKR) ' becomes order_amount_pkr. Much tidier!

Block 6: Filling in the Blanks

```
df_clean[col] = df_clean[col].fillna('Unknown')
df_clean[col] = df_clean[col].fillna(df_clean[col].median())
```

In simple words: Missing text becomes "Unknown" and missing numbers get filled with the middle (median) value, so no box is left empty.

Dry run: A blank region becomes "Unknown". A blank quantity becomes something like 10 (the typical middle value).

Block 7: Making Numbers Fair to Compare

```
df_clean[f'{col}_normalized'] = (df_clean[col] - df_clean[col].min()) /
    (df_clean[col].max() - df_clean[col].min())
```

In simple words: This squishes all numbers into a range between 0 and 1, like converting different test scores into percentages so you can compare them fairly.

Student Name: Humna Imran

Dry run: An order_amount of 25000, somewhere between 500 and 50000, might become 0.49, which sits right in the middle.

Block 8: Splitting Date Ranges

```
split_dates = df_clean['order_date_range'].str.split(' - ', expand=True)
df_clean['start_date'] = pd.to_datetime(split_dates[0])
df_clean['end_date'] = pd.to_datetime(split_dates[1])
```

In simple words: A single messy text like "2023-01-01 - 2023-01-15" gets split into two clean columns: a start date and an end date.

Dry run: "2023-03-01 - 2023-03-15" becomes start_date = 2023-03-01 and end_date = 2023-03-15, with duration_days = 14.

Block 9: Saving the Clean File

```
df_clean.to_csv('cleaned_data.csv', index=False)
```

In simple words: The cleaned, tidy table is saved as a brand-new file, just like handing in your corrected homework.

Dry run: A new file called cleaned_data.csv appears, fully cleaned and ready to use.

Student Name: Humna Imran

Quick Summary

Step	What it does
1 and 2	Get tools ready and make a practice messy table
3	Open the file
4	Count how messy it is
5	Fix the column titles
6	Fill in missing data
7	Make numbers fair to compare
8	Split date ranges into start and end
9	Save the clean version